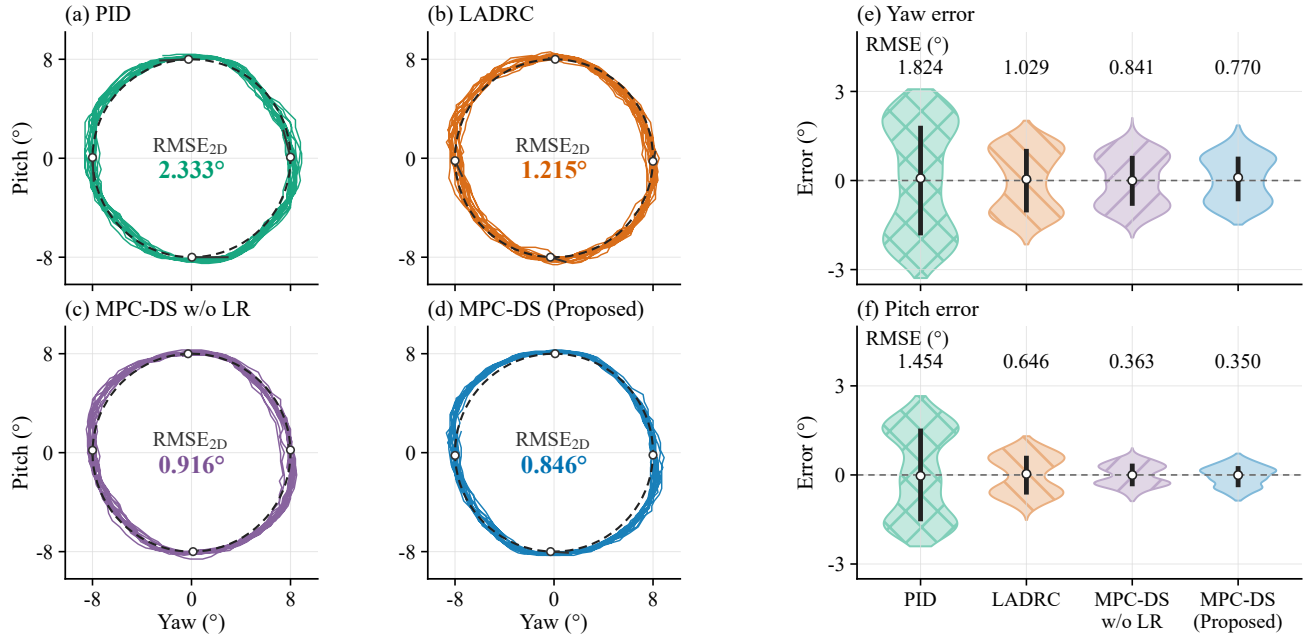

(a) PID
(b) LADRC
(c) MPC-DS w/o LR
(d) MPC-DS (Proposed)
RMSE2D
2.333°
1.215°
0.916°
0.846°
Pitch (°)
Yaw (°)
8
0
-8
Reference
Same-time pair
(e) Yaw error
(f) Pitch error
RMSE (°)
1.824
1.029
0.841
0.770
1.454
0.646
0.363
0.350
Error (°)
3
-3
PID
LADRC
MPC-DS w/o LR
MPC-DS (Proposed)
LR: learned lead residual